

Toka Alokda, M.Sc.

Ph.D. researcher in Cosmology

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Links [LinkedIn](#) - [Github](#) - [Website](#)

Education

University of Bonn (Rheinische Friedrich-Wilhelms-Universität Bonn) - Bonn, DE.

Ph.D. - Astrophysics.

International Max Planck Research School for Astronomy and Astrophysics (IMPRS A&A).

Excellence Cluster “Our Dynamic Universe”.

Topic: Exploring Primordial Non-Gaussianity through the Large Scale Structure with Machine Learning.

Advisor: Prof. Dr. Cristiano Porciani, Dr. Alexander Eggemeier.

June ‘24 - Present

University of Bonn (Rheinische Friedrich-Wilhelms-Universität Bonn) - Bonn, DE.

M.Sc. - Astrophysics.

Bonn-Cologne Graduate School for Physics and Astronomy.

Thesis: “Identification of Protohalos with Deep Learning” (Grade: 1.0¹)

Grade: 1.6 (ranking: 3rd).

Oct. ‘21 - Dec. ‘23

University of Science and Technology at Zewail City - Giza, EG.

B.Sc. - Physics of Earth and Universe; Astrophysics concentration.

GPA: 3.63 ² (Cum Laude).

Sep. ‘17 – Jul. ‘21

Publications

Alokda, T., Porciani, C. and Eggemeier, A. (2026). *Coverage is not enough: Frequentist tests of simulation-based inference for primordial non-Gaussianity*. arXiv: 2605.00980

Alokda, T. and Porciani, C. (2025). *Segmenting proto-halos with vision transformers*. JCAP 11 (2025) 083.

Awards & Honors

Bonn-Cologne Graduate School of Physics and Astronomy (BCGS) Scholarship

Oct. ‘21 - Sep. ‘23

Zewail City UST Scholarship

Sep. ‘17 - Jul. ‘21

B.Sc. Honors

¹ Indicated in the German system; whereas 1.0 is the highest grade and 5.0 is the lowest.

² Indicated in the American system; whereas 4.0 is the highest grade and 0.0 is the lowest.

- **Cum Laude** - Awarded for being among the top students in the graduating class of 2021.
- **Provost's Honors Roll** - Awarded on a semesterly basis for being among the top students.

Jul. '21

Scientific Presentations

"Coverage is not enough: What SBI posteriors of f_{NL} are actually telling you"

Waterloo, CA - Contributed talk in the "Advancing simulation-based and field-level inference for Cosmology" conference. Jun. '26

"The quest for primordial non-Gaussianity: Opportunities and Challenges"

Sofia, BG - Contributed lightning talk at the CosmoVerse 2026 workshop. May '26

"Segmenting Protohalos with Vision Transformers"

Cologne, DE - Contributed talk in the Astroinformatics 2026 conference. Feb. '26

"Identification of Protohalos with Deep Learning"

Les Houches, FR - Contributed talk at the Les Houches - Dark Universe school. Jun. '25

"Identification of Protohalos with Deep Learning"

Cambridge, MA - Online poster presentation in the first AstroAI workshop. Jun. '24

"Identification of Protohalos with Deep Learning"

Bonn, DE - Poster presentation in the annual BCGS poster session. Apr. '24

"CNNs and ViTs for Cosmological Structure Formation"

Bonn, DE - Invited talk in the "Machine Learning Seminar". Dec. '23

"Identification of Protohalos with Deep Learning"

Bonn, DE - Master's colloquium at AIfA, Bonn. Nov. '23

Technical skills

- **Programming & Scientific Computing:**

Python (advanced), C++ (advanced), Julia (intermediate), MATLAB (intermediate), Wolfram Mathematica (intermediate), Fortran (basic).

- **Machine Learning & Differentiable Programming:**

PyTorch, TensorFlow, JAX, Keras, Optuna, Kymatio, SciPy.

- **Astrophysics & Cosmology Tools:**

GADGET-4, L-Galaxies, AHF, CLASS, PINOCCHIO, Python libraries for cosmological analysis: e.g. Pylians, PySpectrum, Nbodykit.

- **High-Performance Computing & OS:** Linux, Bash/UNIX shell, Slurm

- **Documentation & Reporting:** LaTeX, MS Office

Teaching Experience

University of Bonn

Oct. '22 - Present

Graduate Teaching Assistant

Summer Semester 26

The large scale structure as revealed by redshift surveys (M.Sc. of Astrophysics)

Winter Semester 22/23, 23/24, 24/25, and 25/26

Maths for Theoretical Astrophysics and Cosmology (M.Sc. of Astrophysics).

Summer Semester 24 and 25

Advanced Topics in Cosmology (M.Sc. of Astrophysics).

Summer Semester 23

Astrophysics of Galaxies (M.Sc. of Astrophysics).

University of Science and Technology at Zewail City

Sep. '20 - Jan. '21

Undergraduate Teaching Assistant

Fall 2021: Quantum Mechanics I (B.Sc. of Physics).

Public Outreach

“Tangled in the Web: The rise of AI in Astronomy” - Astronomy on Tap - Bonn, DE

Oct. '24